

The Geometry of Alignment Collapse: When Fine-Tuning Breaks Safety

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Fine-tuning aligned language models on entirely benign tasks (e.g. math tutoring) unpredictably degrades safety guardrails, even when training data contains no harmful content and developers have no adversarial intent. We show that the prevailing explanation, that fine-tuning updates should be orthogonal to safety-critical directions in high-dimensional parameter space, offers false reassurance: we show that this orthogonality is structurally unstable and collapses under the very dynamics of gradient descent. We then resolve this through a novel geometric analysis, proving that alignment concentrates in low-dimensional subspaces with sharp curvature, creating a brittle structure that first-order methods cannot detect or defend. While initial fine-tuning updates may indeed avoid these subspaces, the curvature of the fine-tuning loss generates second-order acceleration that systematically steers trajectories into alignment-sensitive regions—an effect invisible to all existing defenses. We formalize this mechanism through the **Alignment Instability Condition (AIC)**, three geometric properties that, when jointly satisfied, lead to safety degradation. Our main result establishes a quartic scaling law: alignment loss grows with the fourth power of training time, governed by the sharpness of alignment geometry and the strength of curvature coupling between the fine-tuning task and safety-critical parameters. These results expose a structural blind spot in the current safety paradigm. The dominant approaches to safe fine-tuning of null-space projections, gradient filtering, and first-order constraints address only the initial snapshot of a fundamentally dynamic problem. Alignment fragility is not a bug to be patched; it is an intrinsic geometric property of gradient descent on curved manifolds. Our results motivate the development of curvature-aware methods, and we hope will further enable a shift in alignment safety analysis from reactive red-teaming to predictive diagnostics for open-weight model deployment.

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1 Introduction

Fine-tuning a language model on a dataset of math problems breaks its safety guardrails [12]. Training on creative writing examples erodes medical advice safeguards [1, 64]. Adapting for code generation compromises refusal mechanisms [6]. These are not hypothetical risks—they are documented, reproducible phenomena that occur even when fine-tuning datasets contain no harmful content and developers have no adversarial intent [6, 24, 25, 44]. The central puzzle of alignment preservation is thus:

Why does adapting a model for one capability seemingly unpredictably degrade unrelated safety properties?

The question is urgent because fine-tuning has become the dominant paradigm for deploying language models. In domains where technical expertise, compute resources, and specialized data are limited, fine-tuning open weight models represents the primary path to leverage AI capabilities for practitioners. The ecosystem of open-weight models has expanded rapidly [7, 9, 10, 31, 40], enabling innovation but also creating vulnerability. Current defenses are largely reactive, targeting specific attack patterns [18] or imposing first-order gradient constraints [16, 39, 65]

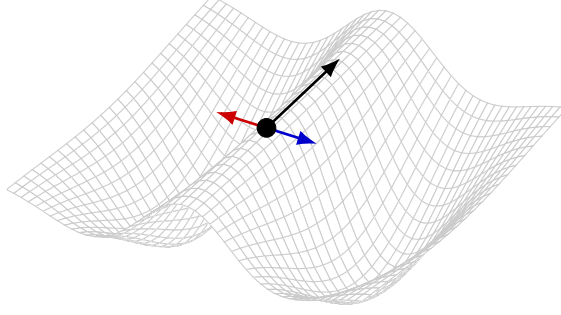


Figure 1: Local alignment instability under fine-tuning dynamics. An illustrative loss landscape showing a fine-tuning trajectory (black arrow) evolving near a ridge separating basins of low alignment utility. Although the initial gradient direction is nearly tangent to the ridge, induced acceleration toward a high-curvature alignment-sensitive direction (red/blue arrow), leads to departure.

that easily break even without adversarial intent [26, 63]. Without understanding the underlying mechanisms of alignment degradation, we cannot build robust safeguards.

The Paradox of High-Dimensional Orthogonality. Empirical evidence reveals a striking pattern: alignment appears to reside in a small subspace of the parameter space. [60] demonstrates that pruning just 3% of specific weights triggers catastrophic safety collapse, while removing similar proportions of weights from other parts of the model leaves its capabilities intact. We call these directions *alignment-sensitive*: parameter updates along them degrade safety, while updates in orthogonal directions should preserve it. This suggests alignment is governed by a low-dimensional set of sensitive directions. In a model with billions of parameters, standard geometric intuition predicts that fine-tuning updates for unrelated tasks should be nearly orthogonal to these directions, thus, safe [5, 35]. Yet degradation occurs frequently across diverse tasks and model families, exhibiting consistent patterns [6, 25, 44].

Existing theoretical frameworks do not fully resolve this paradox. Approaches based on data overlap [25, 26, 63] cannot fully explain why superficially similar tasks have vastly different safety impacts. First-order geometric constraints assume static sensitivity subspaces and fail to account for how fine-tuning trajectories evolve over time [5, 45]. Null-space methods [39, 65] presuppose we can identify and avoid alignment-sensitive directions but offer no explanation for why avoidance fails in practice.

Our Contributions. We resolve this paradox by developing a theory of alignment degradation grounded in the curvature of trajectories in parameter space. Our key insight is that while initial fine-tuning updates may indeed be orthogonal to alignment-sensitive directions, this orthogonality is unstable under the dynamics of gradient descent. The curvature of the fine-tuning objective—captured by second-order terms in the loss landscape—systematically bends trajectories into alignment-sensitive subspaces, even when first-order updates point away from them.

We formalize this mechanism through the **Alignment Instability Condition (AIC)**, which identifies three interacting structural properties arising from the model’s geometry and the fine-tuning task.

1. **Low-Rank Sensitivity:** alignment for a given skill (or capability) concentrates in a small subspace with sharp curvature, characterized by the leading eigenvectors of the Fisher information matrix.
2. **Initial Orthogonality:** fine-tuning gradients for unrelated tasks have negligible first-order projection onto this subspace, creating the illusion of safety.
3. **Curvature Coupling:** second-order terms in the gradient flow induce nontrivial acceleration into the alignment-sensitive subspace, forcing trajectories to drift despite initial orthogonality.

Under the AIC, we prove that alignment degradation is not merely possible, but structurally inevitable within the trust region of the aligned model. Our main theoretical results are:

Theorem 1.1 (Informal Version of Theorem 6.1). *Movement in the alignment-sensitive subspace $M_i \subseteq \mathbb{R}^n$ incurs quadratic utility loss, $\Omega(\lambda \|P_i(\Delta\theta)\|^2)$ where λ is the minimum curvature in M_i and P_i projects onto this subspace.*

Theorem 1.2 (Informal Version of Theorem 6.2). *Along gradient flow trajectories, $\theta(t)$ where $\theta(0) = \theta^*$, satisfying the AIC, projection on M_i grows quadratically: $\|P_i(\theta(t) - \theta^*)\| = \Omega(\gamma t^2)$, where γ quantifies curvature coupling.*

The two main theorems combine to yield a quartic scaling law for alignment degradation within our utility-based framework (defined in Section 3).

Corollary 1.3 (Informal Version of Corollary 6.3). *Combining geometric loss with quadratic drift yields alignment degradation scaling as $\Omega(\lambda \gamma^2 t^4)$ in early training.*

The quartic scaling law provides a principled explanation for rapid, seemingly sudden safety failures observed in practice [20, 42, 43, 60]. Even small amounts of curvature coupling (γ) compound dramatically through the fourth power, causing trajectories to escape safety basins faster than first-order analysis would predict.

Implications. Our theory reveals that alignment fragility is not an artifact of adversarial attacks or fixable through better data curation. Rather, it is an intrinsic geometric property of fine-tuning in high-dimensions. The AIC explains why: (1) benign tasks break safety, (2) low-rank adaptations fail, and (3) first-order defenses are insufficient. Our results necessitate a fundamental rethinking of alignment preservation strategies. In Section 7, we empirically validate these predictions, showing that our notion of geometric overlap provides a principled explanation for degradation of alignment.

2 Related Work

Our work is positioned at the intersection of several research domains: the geometric analysis of neural network parameter spaces, the empirical study of alignment fragility, and the theoretical frameworks of multi-task and meta-learning. We here discuss related studies which were not covered in the main text.

Geometric and Manifold-Based Approaches. The conceptualization of neural network representations and parameter spaces through a geometric lens is well-established [2, 47, 54]. This line of inquiry is broadly motivated by the manifold hypothesis [46], which posits that high-dimensional data, such as inputs for unsupervised learning or distinct classes for classification, concentrates near low-dimensional manifolds separated by regions of low density. More recently, this geometric perspective has been directly applied to the process of fine-tuning [38]. Specifically, Low-Rank Adaptation (LoRA) has been theoretically framed as a problem of manifold geometry [29]. This aligns with foundational work in numerical optimization that analyzes fixed-rank matrices as an embedded geometry, even outside the context of model fine-tuning [46].

While these approaches provide valuable intuition, they often treat alignment geometry as static. Recent work has attempted to identify null-spaces to the alignment weight subspace in an effort to constrain fine-tuning [65]. However, these methods typically assume a flat local geometry that does not account for the empirical fragility of alignment in high-dimensions—a gap our work addresses by integrating this geometric view with a probabilistic analysis of curvature-induced drift.

Alignment Drift and Critical Parameters. Empirical research confirms that fine-tuning modifies only a small subset of the representational subspace, presumably preserving the pre-trained model’s broad capabilities. However, this preservation is asymmetric: while general capabilities may persist, aligned guardrails are uniquely brittle. [26] demonstrate that this degradation is systematic, showing that the impact of fine-tuning varies dramatically depending on the specific task structure, a phenomenon further categorized in broader surveys of trustworthy AI [36]. Critically, this degradation occurs even on seemingly benign tasks such as solving math word problems [12] or standard instruction following [24, 25, 44] where seemingly no adversarial intent exists.

More critically, [16] introduce the “DriftCheck” dataset, demonstrating that even minor LoRA updates induce significant alignment drift. In fine-tuning with this dataset, the identify alignment critical parameters that are most relevant to maintaining safety behavior, finding that an overlap between the fine-tuning data and safety distributions accelerates this drift by overwriting fragile weight regions. This observation connects directly with the safety relevant weights hypothesis of [60], which assesses the brittleness of safety alignment via pruning, showing that removing specific weights triggers rapid safety collapse. [25] and [24] further demonstrate that fine-tuning on benign data can compromise safety even without adversarial intent. Our proposal seeks to explore the theoretical mechanism, which